

SLE251 Research Methods and Data Analysis

PRACTICAL WORKSHEET 4: Correlation and regression

This week we will look at tests that examine the relationship between two continuous (i.e. numeric, or quantitative) variables. The first test is a correlation test: which simply examines whether there is a significant correlation between the two variables (e.g. does one variable increase as the other increases?) Correlation makes no assumption about which one of the two variables is the predictor or response variable. The second test is linear regression, which DOES make an assumption in regard to which one of the variables is the predictor, and which is the response. Regression describes the exact form of the relationship between the two variables, in the form of a regression equation consisting of a slope estimate and an intercept, and can tell you how much the response variable changes in response to a given change in the predictor variable. It can also reveal the extent to which the variation in the response variables is explained by the predictor variable. Like correlation, regression can also more generally reveal if there is a significant relationship between the two variables.

As with the previous worksheets the blue boxes show you a sample of the datasets that you need to load into R and the black boxes will instruct you on the actions you need to do within R to perform the required calculations.

By the end of this worksheet, you will be able to:

- a) carry out a simple correlation and regression analysis in R commander
- b) Interpret the output of these analyses, and associated tests, in order to draw conclusions about the nature of the relationship between two continuous variables
- c) Perform the non-parametric version of a correlation – Spearman's rank correlation

Question 1 – calculating correlation and regression

Diamond et al (2014) were investigating the locomotor performance of curly-tailed lizards (*Leiocephalus carinatus*), and wished to examine their endurance levels. They caught 45 lizards from the wild, and examined their performance on a treadmill (moving at 0.66km/h). They measured endurance as the time (in seconds) the animal spent on the treadmill before it was unable to continue running, and fell off the back of the treadmill. They were interested to see whether larger curly-tailed lizards had greater endurance levels than smaller curly-tailed lizards. The data are in the file *SLE251curlylizards*

Format of *sle251curlylizards* data file

SVL	Endurance
61	36
75	34
75	65
78	33
79	71
79	58

SVL Snout-Vent Length (mm) – a measure of body size of the lizard

Endurance Time (in seconds) spent on the treadmill

- a) First, do a scatterplot of the two variables, with boxplots on both axes, to assess the assumptions of normal distributions and a linear relationship. Paste your plot below. Note: to add the your R plot to the worksheet simply save the graph as a pdf or jpeg and then “Insert” into the word document below.

To create a scatter plot in R

Graphs

Scatterplot

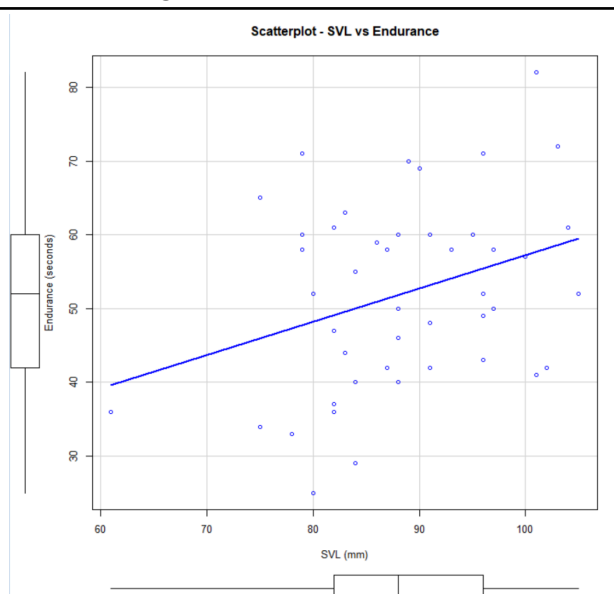
Select *x-variable* (X-axis) and *y-variable* (Y-axis) – use *SVL* as x-axis and *Endurance* as y-axis

Click on the Options Tab

Unselect *Smooth line* and *show spread*

Keep *Marginal boxplots* and *Least-squares line* checked

OK



- b) What do you infer about whether the data for the two variables are normally-distributed or not?
 - The data is normally distributed
- c) Is there any evidence for non-linearity in the relationship between the two variables?
 - Linearity can be observed in the relationship between the 2 variables, evidenced by an equal distribution of data along the median line.

Now test the correlation between the two variables.

To run a correlation in R

Statistics

Summaries

Correlation test...

Choose *SVL* and *Endurance* as the two variables to correlate.

Choose *Pearson* (parametric) correlation co-efficient under *Type of Correlation*.

Leave *Alternative Hypothesis* as *Two-sided*

OK

- d) What is the null hypothesis of this correlation test?
 - There is no correlation between body size and endurance
- e) What is the value of the correlation coefficient (labeled 'cor' in the R output)?
 - $cor = 0.3261769$
- f) What is the *P* value?
 - $p = 0.02877$
 - as less than 0.05, we can reject null hypothesis
- g) What conclusions would you draw from this result?
 - From this result we can derive a positive correlation.

We might be interested in predicting what the endurance capabilities of a lizard are, based on its body size. This requires a linear regression analysis with body size (snout-vent length) as the predictor (*X*) variable and endurance as the response (*Y*) variable.

- h) What is the **main null hypothesis**, associated with the slope, that is being tested using a linear regression analysis?
 - The slope is equal to 0, this means there is no regression between the 2 variables.

- i) In the table below, describe how the assumptions of simple linear regression are checked.

Assumption	How is this checked?
Normally distributed response variable	boxplot or scatterplot
Equal variance in response variable for different values of predictor variable	levene's test
Linear relationship between response and predictor variable	observe trend in scatterplot
Observations independent of each other	random selection

Now fit the regression model $\text{Endurance} = \text{intercept} + \text{slope} \times \text{SVL}$.

To run a linear regression in R

Statistics

Fit models

Linear regression...

You can enter a name for the results object (*Enter name for model:*) but its simplest to just use the name that R provides.

Select *Endurance* from *Response variable* list

Select *SVL* from *Explanatory variables* list.

OK

```
Call:
lm(formula = MPG.city ~ Min.Price)

Residuals:
    Min       1Q   Median       3Q      Max
-6.6505 -1.7971 -0.4599  1.7501  7.2069

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  26.58176    1.02976   25.814 < 2e-16 ***
Min.Price   -0.34008    0.05643   -6.027 2.64e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.018 on 46 degrees of freedom
Multiple R-squared:  0.4412, Adjusted R-squared:  0.4291
F-statistic: 36.32 on 1 and 46 DF, p-value: 2.64e-07
```

The equation of the fitted regression line is:
 $\hat{y} = 26.58 - 0.34x$

- j) Examine the regression output and identify the following:

Sample y-intercept of regression line

Value (*Intercept Estimate* in the R output): 12.0120

Sample slope of regression line (SVL)

Value (*SVL Estimate* in the R output):

Interpretation: 0.4519

t statistic associated with the slope estimate

Value: 2.263

P-value associated with the slope H_0 (regression slope equals zero)

Value: 0.0288

Interpretation: The p-value of 0.0288 suggests that there is evidence to reject the null hypothesis, indicating a statistically significant linear relationship between the

predictor variable(s) and the response variable, with the regression slope being different from zero.

R² value (multiple R-squared)

Value: 0.1064

Interpretation: The R² value of 0.1064 indicates that approximately 10.64% of the variability in the response variable can be explained by the linear relationship with the predictor variable(s).

Carry out an ANOVA for the regression analysis

To create ANOVA table in R

After doing original regression analysis

Models

Hypothesis tests

ANOVA table

Select *Partial, ignoring marginality* ("Type III").

OK

k) Complete the table below - ignore the line in the output that says (Intercept)

Source of variation	df	F-value	P-value
SVL	1	0.4581	0.02877
Residual	43		

l) What biological conclusion would you draw from the analysis?

-
-
-

Question 2 – using regression for prediction

The ability to calculate the mass of the root system of trees is a really important aspect of understanding the amount of stored carbon in forests. However, for obvious reasons, actually measuring the root system of individual trees is impossible without digging up the roots and hence killing the tree. Clearly, on a large scale, this would be inappropriate. Lai and colleagues (2013) sought to investigate whether another, much easier measure to obtain, tree diameter (diameter at breast height – DBH) could serve as a proxy for root biomass, and be used to calculate root biomass for the tree. They went to a pine plantation that was being thinned, and extracted the root system for 39 trees which had been removed, after first measuring the diameter of the tree. Their data can be found in the file *SLE251roots*

Format of *sle251roots* data file

DBH	rootmass
101.4	15.053
17.3	0.379
91.3	12.412
21.8	1.645
45.7	6.177
..	..

DBH Diameter of the tree at Breast Height (cm)

Predictor variable

rootmass Mass of the root system of the tree (kg)

Response variable

- a) What is the main null hypothesis for the linear regression of the slope of rootmass on DBH?
- The slope is equal to 0, this means there is no regression between the 2 variables.

- b) Draw a scatterplot of *rootmass* against *DBH*, with marginal boxplots for each variable as well. Is there any evidence of skewness in the distributions or nonlinearity (to visualize the latter check the 'Smooth line' option under the options tab of the scatterplot)? Paste the scatterplot below (with least-squares line and smoothed line both displayed)

To create a scatter plot in R

Graphs

Scatterplot

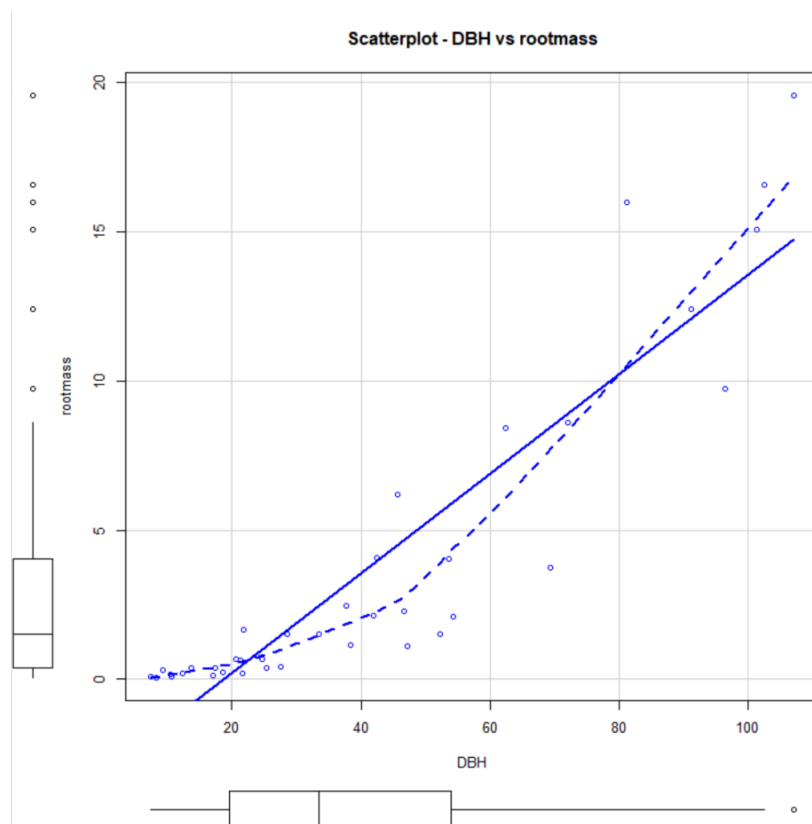
Select *x-variable* (X-axis) and *y-variable* (Y-axis) – use *DBH* as the x-axis and *rootmass* as the y-axis

Click on the Options Tab

Unselect *Show spread*

Keep *Marginal boxplots*, *Least-squares line* and *smooth line* checked

OK



- not normally distributed and not linear, observed by dip in dotted line
- note that graph data is skewed for both variables with significant outliers

- c) $\log_{10}$ transform both variables (*DBH* and *rootmass*). For the log-transformed data, redo the scatterplot with marginal boxplots and with both the least-squares and smoothed line displayed – and paste below.

To log transform data in R

Data

Manage variables in active data set

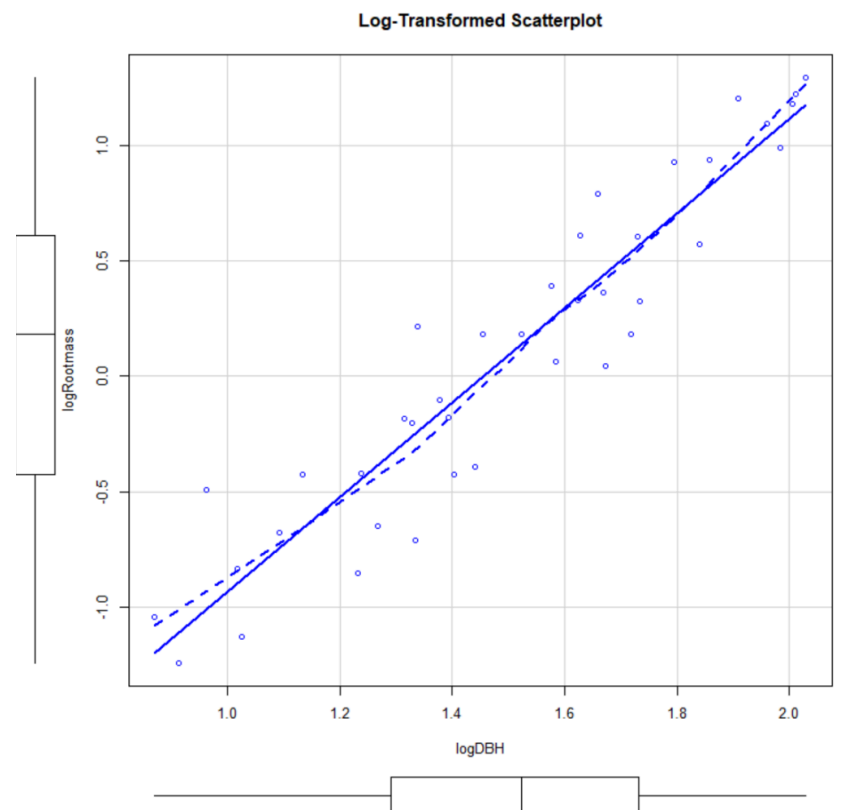
Compute new variable

Type in name of new transformed variable (e.g. *logDBH*) in *New variable name* box (always create a new variable for the transformed data)

Type in transformation required in *Expression to compute* box. For example if *DBH* is the name of the variable to be transformed, the expression *log10(DBH)* would be the appropriate one to type

OK

Repeat for the rootmass variable



- log transformed data appears more normally distributed and linear
- data is no longer skewed and does not have significant outliers

- d) Describe the differences as a result of log transformation. Are the data more normally distributed and do they have better linearity?

Now fit the regression model $\log\text{rootmass} = \text{intercept} + \text{slope} \times \log\text{DBH}$

NB: USE THE LOG TRANSFORMED DATA FOR THIS ANALYSIS

To fit linear regression in R

Statistics

Fit models

Linear regression...

You can enter a name for the results object (*Enter name for model:*) but its simplest to just use the name that R provides.

Select $\log\text{rootmass}(*)$ from *Response variable* list (* or whatever you called this logged variable)

Select $\log\text{DBH}$ from *Explanatory variables* list.

OK

- e) Write the results out as though you were writing a research paper/thesis (delete the phrases which do not apply and fill in gaps with your results):

The linear regression of log root biomass against log DBH showed a

Significant positive relationship

Significant negative relationship

No significant relationship

between the two variables ($\text{Slope} = 2.0461$, $\text{std.error} = 2.0461$, $t = 17.09$, $P = < 2.2\text{e-}16$).

- f) Write out the regression equation (fill in intercept and slope)

$$\log\text{rootmass} = (\quad) + (\quad) * \log\text{DBH}$$

$$\log\text{rootmass} = \text{intercept} + \text{slope} * \log\text{DBH}$$

- g) What conclusion would you draw from the regression analysis? Could DBH be used as a predictor of root bio mass?
- h) What percentage of the variation in log rootmass is explained by the linear relationship with log DBH (use the r^2 value)?
- i) Using the regression equation above, calculate what root mass a tree with a diameter at breast height of 150cm would have. THINK CAREFULLY – you need to convert 150 into the log scale - use the excel function **LOG10(150)**. Then plug value this into the regression equation, and finally convert the answer for logrootmass back into the original scale – use the excel function **Power(10,y)** – where y is the value for logrootmass you got from the regression equation.
- Rootmass (in kg) for a pine tree with DBH of 150cm =

Question 3 – Non-parametric correlation (Spearman's)

A scientist at a local tennis club wished to investigate whether there was any association between natural levels of testosterone in the blood and success in a men's tennis league at the club. At the end of the season they took blood samples from the 16 men who were in the league and related their testosterone level to their final position on the ladder. The results are shown in the file *SLE251tennis*

Format of *sle251tennis* data file

Tableposition	Testosterone
1	664.88
2	468.33
3	765.34
4	889.85
5	805.09
..	..

Tableposition Position of the tennis player in the league at the end of the season

Testosterone Levels of testosterone in blood (ng/dl)

- What is the null hypothesis of this correlation test?
 - There is no correlation between table position and testosterone
- Explain why a non-parametric Spearman's correlation (and not the parametric Pearson's correlation test) is the appropriate correlation test to run in this case (hint: think about the nature of the Tableposition variable)

-

Now test the Spearman's correlation between the two variables.

To run a correlation in R

Statistics

Summaries

Correlation test...

Choose *Tableposition* and *Testosterone* as the two variables to correlate.

Choose the *Spearman* co-efficient under *Type of Correlation*.

Leave *Alternative Hypothesis* as *Two-sided*

OK

NOTE: The correlation coefficient in this test is given under the value 'rho'

- Write the results out as though you were writing a research paper/thesis (delete the phrases which do not apply and fill in gaps with your results):

There was

No significant correlation

between position in the tennis league and testosterone level ($r_s = 0.02058824$, $P = 0.9432$).

- What conclusion would you draw from this analysis?
 - There is no significant correlation between the 2 variables.